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Event-triggered control for boundary controlled time-fractional diffusion systems with spatially-varying coefficients

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ABSTRACT

In this paper, an event-triggered control scheme is proposed for boundary controlled time-fractional diffusion systems with a semilinear function and the spatially-varying coefficients. Toward this aim, backstepping method is utilized. The use of backstepping allows us to explicitly express our proposed event-triggered boundary controllers. Specifically, we first discuss well-posedness of the solution to the problem under consideration. An event-triggered boundary controller is then designed using backstepping to Mittag-Leffler stabilize the system under consideration. After these, we show that our event-triggered controller can efficiently prevent the existence of so-called Zeno phenomenon. Finally, we give a numerical example to illustrate the effectiveness of designed event-triggered boundary control scheme.

1. Introduction

Stabilization problem for diffusion systems described by partial differential equations (PDEs) of parabolic type is a basic issue in applied mathematics. Over the past two decades, a considerable number of control strategies, such as adaptive control [1], event-triggered control [2] and continuous-time control [3] have been developed to address the stabilization problem of diffusion systems. In the midst of them, event-triggered control can obtain a desired stabilization performance under a reduced stabilizer updating and information transmissions, hence saving both energy and computation resources [4,5]. With these, numerous studies on event-triggered control problems of lumped parameter systems governed by ordinary differential equations (ODEs), such as linear ODE systems [6], nonlinear ODE systems [7,8], stochastic systems [9,10] and semi-Markovian jump systems [11,12], etc., have appeared. Moreover, in recent years, the study of event-triggered control has been extended to diffusion PDE systems and many meaningful works were obtained. For example, in [13], event-triggered control for a class of linear diffusion PDEs was first investigated by using the model reduction method. Subsequently, event-triggered in-domain control problems for semilinear diffusion PDEs and the Korteweg-de Vries equation were studied via Lyapunov stability theory in [14] and [15], respectively. The authors in [16] discussed fuzzy adaptive control of semilinear diffusion PDE systems via distributed event-triggered mechanism.

It should be mentioned that the aforementioned studies mainly focus on traditional integer-order diffusion systems and many components in these studies, which can influence the structure of the model such as the nonlocal and hereditary properties of

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considered complex diffusion processes, are ignored [17]. For dealing with this, as is stated in [18,19], the nonlocal and hereditary properties of complex diffusion processes need to utilize time-fractional diffusion systems to achieve a more accurate description. Here time-fractional diffusion system is obtained by changing the time derivative of traditional diffusion systems to a fractional-order derivative. Notice that boundary control is a fundamental way for controller to act on time-fractional diffusion systems, whose actuators are only located at system domain's boundary that is easily to be carried out. Within this context, it would be of great significance to do profound research on the event-triggered control of boundary controlled time-fractional diffusion systems.

However, owing to the fact that fractional calculus and stability analysis of time-fractional diffusion systems certainly challenge the studies, it is only in the last several years that a few studies have been gained in the literatures on event-triggered boundary control problems of various time-fractional diffusion systems. In [20], event-triggered state feedback and output feedback controller design for a boundary controlled linear time-fractional diffusion system with input uncertainties were considered. The authors in [21] dealt with the boundary consensus control problems of time-fractional multi-agent systems with diffusion terms. And in [22], based on fractional Lyapunov direct method and the linear matrix inequalities, the event-triggered consensus control for boundary controlled multi-agent time-fractional reaction-diffusion systems was discussed. Notice that time-fractional diffusion systems with a semilinear function and the spatially-varying coefficients could suffice for a much wider range of real-applications and nowadays, increasing attention has been attracted to the continuous-time control of time-fractional diffusion systems with spatially-varying coefficients [23,24]. Thus, investigation on event-triggered control problem for boundary controlled semilinear time-fractional diffusion systems with spatially-varying coefficients should be both interesting and challenging. To the best knowledge of authors, this problem is still undiscussed, which inspires our present work.

Summarizing the above considerations, herein, we consider the boundary controlled semilinear time-fractional diffusion systems with spatially-varying coefficients as follows:

$$\begin{cases} {}^C_0 D_t^\alpha \omega(x, t) = \beta(x) \frac{\partial^2 \omega(x, t)}{\partial x^2} + \gamma(x) \frac{\partial \omega(x, t)}{\partial x} + f(\omega(x, t), x, t) & \text{in } (0, L) \times [0, \infty), \\ \frac{\partial \omega}{\partial x}(0, t) = p\omega(0, t), \quad \frac{\partial \omega}{\partial x}(L, t) + q\omega(L, t) = u(t) & \text{in } [0, \infty), \\ \omega(x, 0) = \omega_0(x) & \text{in } (0, L), \end{cases} \quad (1.1)$$

where $\omega_0 \in L^2(0, L)$ is the initial condition, $\beta \in C^2[0, L]$, $\gamma \in C[0, L]$ satisfying $0 < \beta_{\min} \leq \beta(x) \leq \beta_{\max}$, $0 \leq \gamma(x) \leq \gamma_{\max}$ are spatially-varying coefficients,

$${}^C_0 D_t^\alpha \omega(\cdot, t) = {}_0 I_t^{1-\alpha} \frac{\partial \omega}{\partial t}(\cdot, t), \quad \alpha \in (0, 1] \quad (1.2)$$

represents the Caputo fractional derivative and ${}_0 I_t^\alpha \omega(\cdot, t) = \int_0^t \frac{(t-s)^{\alpha-1}}{\Gamma(\alpha)} \omega(\cdot, s) ds$ is the fractional-order integral of Riemann-Liouville type [25]. Besides, $u(t)$ denotes the boundary control input, p, q are arbitrary constants ($p = q = +\infty$ yields the Dirichlet boundary conditions (BCs) case) and $f(\omega(x, t), x, t)$ is the semilinear function to be given later. Here $L^2(0, L)$ represents a Hilbert space endowed with the inner product $(\phi_1, \phi_2)_{L^2(0, L)} = \int_0^L \phi_1(x) \phi_2(x) dx$ and the norm $\|\phi_1\|_{L^2(0, L)} = \sqrt{\int_0^L \phi_1^2(x) dx}$. By [17,18], one sees that this type of time-fractional diffusion system plays a significant role for well modeling a rather large class of science and engineering applications, see for example the dynamics of the nuclear reactor [26] and the heating processes of non-uniform metal slabs [20], etc.

The goal of this paper is to investigate event-triggered boundary control problem of semilinear time-fractional diffusion systems (1.1). Based on [23,27], backstepping method will be utilized, which allows us to explicitly express our proposed event-triggered boundary controllers that is very useful for real-applications. Specifically, we first study well-posedness of the studied system (1.1) and then, formulate an event-triggered controller design strategy via backstepping ensuring that the corresponding closed-loop system is Mittag-Leffler stable. For event-triggered control scheme, to show the non-occurrence of so-called Zeno behavior is very important [16]. In this paper, we shall focus on proving that the inter-event time of two neighboring instants for the designed event-triggered control scheme has a lower bound by seriously analyzing the evolution trends of the event-triggered condition.

We proceed as follows. In the next section, we present our problems and introduce several preliminaries that are useful for our studies. Section 3 is devoted to formulating the backstepping-based event-triggered controller design strategy for the considered time-fractional diffusion systems. Besides, we prove that the existence of Zeno behavior can be avoided. This is tested in Section 4, where a numerical experiment is included.

2. Preliminary results

To begin with, we see that based on variable transformation $\omega(x, t) \mapsto \omega(x, t) e^{-\int_0^x \frac{\gamma(s)}{2\beta(s)} ds}$, the term $\gamma(x) \frac{\partial \omega(x, t)}{\partial x}$ in above system (1.1) can be circumvented and moreover,

$$\begin{aligned} f(\omega(x, t), x, t) &\mapsto f\left(\omega(x, t) e^{-\int_0^x \frac{\gamma(s)}{2\beta(s)} ds}, x, t\right) e^{\int_0^x \frac{\gamma(s)}{2\beta(s)} ds} - \frac{\gamma^2(x)}{2\beta(x)} \omega(x, t) - \beta(x) e^{\int_0^x \frac{\gamma(s)}{2\beta(s)} ds} \frac{d}{dx} \left(\frac{1}{2} \frac{\gamma(x)/\beta(x)}{e^{\int_0^x \frac{\gamma(s)}{2\beta(s)} ds}} \right) \omega(x, t), \\ p &\mapsto p + \frac{\gamma(0)}{2\beta(0)}, \quad q \mapsto q - \frac{\gamma(L)}{2\beta(L)}, \quad u(t) \mapsto u(t) e^{\int_0^L \frac{\gamma(s)}{2\beta(s)} ds}, \quad \omega_0(x) \mapsto \omega_0(x) e^{-\int_0^x \frac{\gamma(s)}{2\beta(s)} ds}. \end{aligned} \quad (2.1)$$

Note that these additive terms and integral can't affect our analysis and can be easily included. In what follows, we continue to investigate system (1.1) and just simply set $\gamma(x) \frac{\partial \omega(x, t)}{\partial x} \equiv 0$ throughout this paper.

To obtain our results, it is supposed that $f(\omega(x, t), x, t) : C([0, \infty); L^2(0, L)) \times (0, L) \times [0, \infty) \rightarrow L^2(0, L)$ is locally Lipschitz continuous, i.e.,

Assumption A_0 : $f(0, x, t) = 0$ and given some $r > 0$, a positive constant $c = c(r)$ fulfilling

$$\|f(\omega, x, t) - f(\omega^*, x, t)\|_{L^2(0, L)} \leq c \|\omega - \omega^*\|_{L^2(0, L)} \quad (2.2)$$

for any ω, ω^* with $\max\{\|\omega\|_{C([0, \infty); L^2(0, L))}, \|\omega^*\|_{C([0, \infty); L^2(0, L))}\} \leq r$ exists.

As for the operator $(\mathcal{A}\varphi)(x) = -\beta(x)\frac{\partial^2 \varphi}{\partial x^2}$ under

$$\mathcal{D}(\mathcal{A}) = \left\{ \varphi \in L^2(0, L) : \frac{d\varphi(0)}{dx} = p\varphi(0), \frac{d\varphi(L)}{dx} = -q\varphi(L) \right\}, \quad (2.3)$$

by [28,29], the eigenvalue pairing $\{(\lambda_n, \xi_n)\}_{n \geq 1}$ of operator $(\mathcal{A}, \mathcal{D}(\mathcal{A}))$ satisfies that

a) the eigenvalues $\{\lambda_n\}_{n \geq 1}$ is an increasing sequence and $\lim_{n \rightarrow \infty} \lambda_n = \infty$;

b) $\{\xi_n(x)\}_{n \geq 1}$ is a orthonormal basis of the Hilbert space $L^2(0, L)$.

Therefore, if $f(\omega, x, t) > c_0 |\omega|$ holds for some constant $c_0 > 0$ and c_0 is large enough, we see that when $u \equiv 0$, the studied system (1.1) is unstable.

Before finishing this section, we discuss well-posedness of the studied control problem, thereby making the contribution of this paper clear.

For this, by the definition of event-triggered controllers, we let the increasing sequence $\{t_k\}_{k \geq 0}$ with $t_0 = 0$, $\lim_{k \rightarrow \infty} t_k = \infty$ denote the triggering instants and suppose that event-triggered boundary controller with zero-order hold (ZOH) for system (1.1) is of the following form:

$$u(t) = \bar{u}_k \text{ for all } t \in [t_k, t_{k+1}), \quad k = 0, 1, 2, \dots \quad (2.4)$$

This indicates that the considered system (1.1) is a hybrid system, in which boundary control inputs are discrete constants with ZOH. Further, introduce two Sobolev spaces: $H^1_0(0, L)$ and $H^2(0, L)$ [30], we present the following useful result.

Lemma 2.1. [20,31] For some $T > 0$, let $u : [0, T) \rightarrow \mathbf{R}$ be defined as in Eq. (2.4) which is a right continuous function. Then, there exists a unique solution $\omega \in C([0, T]; L^2(0, L)) \cap C((0, T]; H^2(0, L) \cap H^1_0(0, L))$ satisfying ${}_0^C D_t^\alpha \omega \in C((0, T]; L^2(0, L))$ and the system

$$\begin{cases} {}_0^C D_t^\alpha \omega(x, t) = \beta(x) \frac{\partial^2 \omega(x, t)}{\partial x^2} + f(\omega(x, t), x, t) & \text{in } (0, L) \times [0, T], \\ \frac{\partial \omega}{\partial x}(0, t) = p\omega(0, t), \quad \frac{\partial \omega}{\partial x}(L, t) + q\omega(L, t) = u(t) & \text{in } [0, T], \\ \omega(x, 0) = \omega_0(x) & \text{in } (0, L) \end{cases} \quad (2.5)$$

provided that $\omega_0 \in L^2(0, L)$.

It is worth pointing out that the way for proving Lemma 2.1 is very similar to that of Lemma 2.4 in [20] or Theorems 2.1, 2.2 in [31], we omit its details. Based on Lemma 2.1, one can get how the solution of system (1.1) is constructed via a step-by-step evolution and then, have the following deduced result.

Corollary 2.1. Let the increasing sequence $\{t_k\}_{k \geq 0}$ defined above be the triggering time and define $I = [0, \infty) \setminus \{t_k\}_{k \geq 0}$. Suppose that the event-triggered boundary control $u(t)$ is defined as in (2.4) with ZOH. Then, for any $\omega_0 \in L^2(0, L)$, there is a unique solution $\omega \in C(I; L^2(0, L))$ satisfying $\omega(\cdot, t) \in C^2(0, L)$, $\forall t \in [0, \infty)$ and

$$\begin{cases} {}_0^C D_t^\alpha \omega(x, t) = \beta(x) \frac{\partial^2 \omega(x, t)}{\partial x^2} + f(\omega(x, t), x, t) & \text{in } (0, L) \times I, \\ \frac{\partial \omega}{\partial x}(0, t) = p\omega(0, t), \quad \frac{\partial \omega}{\partial x}(L, t) + q\omega(L, t) = u(t) & \text{in } I, \\ \omega(x, 0) = \omega_0(x) & \text{in } (0, L). \end{cases} \quad (2.6)$$

Since Corollary 2.1 can be directly deduced from Lemma 2.1 via a step-by-step evolution, we omit its proof. Based on Corollary 2.1, it is not difficult to get that the solution of system (1.1) is differentiable only in the open interval (t_k, t_{k+1}) , while is continuous for all $t \in [0, \infty)$. Thus, we can claim that our studied event-triggered boundary control problem is well-posed.

3. Main results

This section is devoted to discussing the event-triggered controller design strategy for system (1.1) using backstepping. By assuming that control inputs are implemented under ZOH and $\{t_k\}_{k \geq 0}$ denotes the triggering instants to be determined later, for each time interval $[t_k, t_{k+1})$, one gets that the considered system becomes

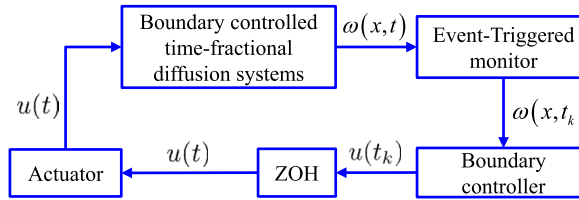


Fig. 1. Diagram for event-triggered control of boundary controlled system (1.1).

$$\begin{cases} {}^C_0 D_t^\alpha \omega(x, t) = \beta(x) \frac{\partial^2 \omega(x, t)}{\partial x^2} + f(\omega(x, t), x, t) & \text{in } (0, L) \times [t_k, t_{k+1}), \\ \frac{\partial \omega}{\partial x}(0, t) = p\omega(0, t), \quad \frac{\partial \omega}{\partial x}(L, t) + q\omega(L, t) = u(t_k) & \text{in } [t_k, t_{k+1}), \\ \omega(x, 0) = \omega_0(x) & \text{in } (0, L). \end{cases} \quad (3.1)$$

3.1. Backstepping transformation

Inspired by available works on using backstepping for traditional diffusion systems and time-fractional diffusion systems, herein, we give the following integral transformation

$$\rho(x, t) = \omega(x, t) - \int_0^x h(x, \varsigma) \omega(\varsigma, t) d\varsigma \quad (3.2)$$

and focus on using it to convert the studied system (3.1) into a target system as follows:

$$\begin{cases} {}^C_0 D_t^\alpha \rho(x, t) = \beta(x) \frac{\partial^2 \rho(x, t)}{\partial x^2} - \mu \rho(x, t) + f(\omega(x, t), x, t) - \int_0^x h(x, \varsigma) f(\omega(\varsigma, t), \varsigma, t) d\varsigma & \text{in } (0, L) \times [t_k, t_{k+1}), \\ \frac{\partial \rho(0, t)}{\partial x} = \hat{p} \rho(0, t) & \text{in } [t_k, t_{k+1}), \\ \frac{\partial \rho(L, t)}{\partial x} + \hat{q} \rho(L, t) = u(t_k) - \hat{q} \int_0^L h(L, \varsigma) \omega(\varsigma, t) d\varsigma - \int_0^L \frac{\partial h(L, \varsigma)}{\partial x} \omega(\varsigma, t) d\varsigma & \text{in } [t_k, t_{k+1}), \\ \rho(x, 0) = \rho_0(x) = \omega_0(x) - \int_0^x h(x, \varsigma) \omega_0(\varsigma) d\varsigma & \text{in } (0, L). \end{cases} \quad (3.3)$$

Here $\mu > 0$ is a designed parameter, $h(x, \varsigma)$ is the kernel of integral transformation (3.2) and $\hat{p}$, $\hat{q}$ are two constants to be specified in the following proposition.

Proposition 3.1. Let $\hat{p} = p$ and $\hat{q} = q - h(L, L)$. For any given constant $\mu > 0$, suppose that $\frac{3\beta'^2(x)}{16\beta(x)} - \frac{\beta''(x)}{4} = C$ is a constant and the kernel $h(x, \varsigma)$ satisfies

$$\begin{cases} \beta(x) \frac{\partial^2 h(x, \varsigma)}{\partial x^2} - \frac{\partial^2 [h(x, \varsigma) \beta(\varsigma)]}{\partial \varsigma^2} = \mu h(x, \varsigma), \\ 2\beta(x) \frac{d}{dx} h(x, x) + h(x, x) \beta'(x) = -\mu, \\ ph(x, 0) \beta(0) - \beta(0) \frac{\partial h(x, 0)}{\partial \varsigma} = h(x, 0) \frac{d\beta(0)}{dx}, \\ h(0, 0) = 0 \end{cases} \quad (3.4)$$

over domain $D = \{(x, \varsigma) : 0 \leq \varsigma \leq x \leq L\}$, then the studied system (3.1) can be equivalently transformed into target system (3.3) by using above integral transformation (3.2).

Proof. See Appendix A. $\square$

Based on Step 1 of the proof of Proposition 3.1, above hyperbolic PDE system (3.4) admits a twice continuously differentiable solution over the domain D . Then, we see that the integral transformation (3.2) is invertible, i.e., (3.2) and its inverse transformation are all bounded. That is, there exists a constant $\nu > 0$ such that

$$\|\omega(\cdot, t)\|_{L^2(0, L)} \leq \nu \|\rho(\cdot, t)\|_{L^2(0, L)}, \quad \|\rho_0\|_{L^2(0, L)} \leq \nu \|\omega_0\|_{L^2(0, L)}. \quad (3.5)$$

3.2. Event-triggered controller design and implementation

As plotted in Fig. 1, our event-triggered control scheme to be proposed is usually made up of two main parts: 1) a monitor that checks if current state information violates a designed triggering condition for determining the triggering instants; 2) a controller that is designed when the proposed triggered condition is violated to Mittag-Leffler stabilize the studied system by a ZOH until the next triggering time occurs.

Toward this aim, we design the event-triggered monitoring condition as follows:

$$t_{k+1} = \min \left\{ t > t_k : \|\omega(\cdot, t) - \omega(\cdot, t_k)\|_{L^2(0,L)} > \check{\epsilon} \|\omega(\cdot, t)\|_{L^2(0,L)} \right\} \quad (3.6)$$

for all $k = 0, 1, 2, \dots$ with $t_0 = 0$, where $\check{\epsilon} > 0$ is a constant denoting the event threshold. With this, one gets that when the condition is triggered, the left part $\|\omega(\cdot, t) - \omega(\cdot, t_k)\|_{L^2(0,L)}$ in (3.6) is reset to zero and starts to grow again until it arrives in next triggering instants. Moreover, as for the event-triggered boundary controller, we propose that $u(t_k)$ is of the following form:

$$u(t_k) = \int_0^L \frac{\partial h(L, \zeta)}{\partial x} \omega(\zeta, t_k) d\zeta + \hat{q} \int_0^L h(L, \zeta) \omega(\zeta, t_k) d\zeta. \quad (3.7)$$

Theorem 3.1. Let assumption A_0 hold. Suppose that all conditions of Proposition 3.1 are satisfied and Laplace transformation of the function $\rho^2(\cdot, t)$ exists. Choose some $r_0 > 0$ such that $\|\omega_0\|_{L^2(0,L)} \leq r_0$ and let the event-triggered boundary control law $u(t_k)$ be defined as in Eq. (3.7). Then, if $\beta(x)$ satisfies

$$\hat{q}\beta(L) + \frac{\beta'(L)}{2} \geq \frac{\beta^2(L)}{4} \geq 0, \quad \hat{p}\beta(0) \geq \frac{\beta'(0)}{2} \quad (3.8)$$

and a $\mu^* > 0$ with respect to $\beta(x)$, $h(x, \zeta)$, $\check{\epsilon}$ can be found such that

$$\mu \geq \mu^* + \varepsilon \quad (3.9)$$

holds true for some $\varepsilon > 0$, system (1.1) with triggering condition (3.6) and the event-triggered boundary controller (3.7) is Mittag-Leffler stable in Hilbert space $L^2(0, L)$ and its solution yields that

$$\sup_{t \geq 0} \|\omega(\cdot, t)\|_{L^2(0,L)} \leq r \text{ for some } r = r(r_0). \quad (3.10)$$

The following lemma is needed for proving Theorem 3.1.

Lemma 3.1. [32] Given any differentiable function $v : [0, \infty) \rightarrow \mathbf{R}$, then

$$\frac{1}{2} {}^C D_t^\alpha v^2(t) \leq v(t) {}^C D_t^\alpha v(t), \quad \alpha \in (0, 1] \quad (3.11)$$

is satisfied for all $t \geq 0$.

Proof of Theorem 3.1. By Proposition 3.1, since systems (3.1) and (3.3) are equivalent under the integral transformation (3.2). Herein, we aim to analyze Mittag-Leffler stability property of target system (3.3).

Choose

$$V(t) = \frac{1}{2} \int_0^L \rho^2(x, t) dx, \quad (3.12)$$

since $\omega(\cdot, t)$ is differential, one gets that the function $\rho(\cdot, t)$ is also differential. So, by the Lemma 3.1, we have

$$\begin{aligned} {}^C D_t^\alpha V(t) &\leq \int_0^L \rho(x, t) {}^C D_t^\alpha \rho(x, t) dx \\ &= -2\mu V(t) + \rho(L, t)\beta(L) \frac{\partial \rho(L, t)}{\partial x} - \rho(0, t)\beta(0) \frac{\partial \rho(0, t)}{\partial x} - \int_0^L \beta(x) \left(\frac{\partial \rho(x, t)}{\partial x} \right)^2 dx \\ &\quad - \int_0^L \frac{\partial \rho(x, t)}{\partial x} \beta'(x) \rho(x, t) dx + \int_0^L \rho(x, t) f(\omega(x, t), x, t) dx \\ &\quad - \int_0^L \rho(x, t) \int_0^x h(x, \zeta) f(\omega(\zeta, t), \zeta, t) d\zeta dx. \end{aligned} \quad (3.13)$$

According to BCs of system (3.3), it then follows that $\rho(0, t)\beta(0) \frac{\partial \rho(0, t)}{\partial x} = \hat{p}\beta(0)\rho^2(0, t)$ and

$$\rho(L, t)\beta(L) \frac{\partial \rho(L, t)}{\partial x} = -\hat{q}\beta(L)\rho^2(L, t) + \beta(L)\rho(L, t) \left(u(t_k) - \hat{q} \int_0^L h(L, \zeta) \omega(\zeta, t) d\zeta - \int_0^L \frac{\partial h(L, \zeta)}{\partial x} \omega(\zeta, t) d\zeta \right). \quad (3.14)$$

Notice that the kernel function $h(x, \zeta)$ is twice continuously differentiable, based on the designed event-triggered controller (3.7), it leads to

$$\begin{aligned} &\left| u(t_k) - \hat{q} \int_0^L h(L, \zeta) \omega(\zeta, t) d\zeta - \int_0^L \frac{\partial h(L, \zeta)}{\partial x} \omega(\zeta, t) d\zeta \right| \\ &\leq \left| \int_0^L \frac{\partial h(L, \zeta)}{\partial x} (\omega(\zeta, t_k) - \omega(\zeta, t)) d\zeta \right| + \left| \hat{q} \int_0^L h(L, \zeta) (\omega(\zeta, t_k) - \omega(\zeta, t)) d\zeta \right| \\ &\leq C^* \|\omega(\cdot, t) - \omega(\cdot, t_k)\|_{L^2(0,L)}, \end{aligned} \quad (3.15)$$

where $C^* = \left\| \frac{\partial h}{\partial x}(L, \cdot) \right\|_{L^2(0,L)} + |\hat{q}| \|h(L, \cdot)\|_{L^2(0,L)}$. Again utilizing Assumption A_0 , one has

$$\begin{aligned}
& \int_0^L \rho(x, t) \left(f(\omega(x, t), x, t) - \int_0^x h(x, \zeta) f(\omega(\zeta, t), \zeta, t) d\zeta \right) dx \\
& \leq \left(1 + \frac{C_h L^{\frac{3}{2}}}{\sqrt{3}} \right) \|f(\omega(\cdot, t), \cdot, t)\|_{L^2(0, L)} \|\rho(\cdot, t)\|_{L^2(0, L)} \\
& \leq 2c\nu \left(1 + \frac{C_h L^{\frac{3}{2}}}{\sqrt{3}} \right) V(t)
\end{aligned} \tag{3.16}$$

by following the inequality (3.5) in the section of Appendix and the Cauchy-Schwartz inequality. Here $C_h = \sup_{0 \leq \xi \leq x \leq L} |h(x, \xi)|$ is a constant. It therefore, follows from

$$\int_0^L \beta'(x) \rho(x, t) d\rho(x, t) = \beta'(x) \rho^2(x, t) \Big|_{x=0}^{x=L} - \int_0^L \rho(x, t) \left(\beta'(x) \frac{\partial \rho(x, t)}{\partial x} + \beta''(x) \rho(x, t) \right) dx \tag{3.17}$$

that

$$\int_0^L \frac{\partial \rho(x, t)}{\partial x} \beta'(x) \rho(x, t) dx = \frac{\beta'(L)}{2} \rho^2(L, t) - \frac{\beta'(0)}{2} \rho^2(0, t) - \frac{1}{2} \int_0^L \beta''(x) \rho^2(x, t) dx. \tag{3.18}$$

With these, by the inequalities (3.8), the triggering condition (3.6) and the inequality $0 < \beta_{\min} \leq \beta(x) \leq \beta_{\max}$ on $\beta(x)$, we obtain

$$\begin{aligned}
{}_0^C D_t^\alpha V(t) & \leq - \left(\hat{q}\beta(L) + \frac{\beta'(L)}{2} - \frac{\beta^2(L)}{4} \right) \rho^2(L, t) - 2\mu V(t) \\
& \quad + \frac{1}{2} \int_0^L \beta''(x) \rho^2(x, t) dx - \left(\hat{p}\beta(0) - \frac{\beta'(0)}{2} \right) \rho^2(0, t) \\
& \quad + C^{*2} \|\omega(\cdot, t) - \omega(\cdot, t_k)\|_{L^2(0, L)}^2 + 2c\nu \left(1 + \frac{C_h L^{\frac{3}{2}}}{\sqrt{3}} \right) V(t) \\
& \leq - \left(2\mu - \max_{x \in [0, L]} \beta''(x) - 2C^{*2} \check{\epsilon}^2 \nu^2 - 2c\nu \left(1 + \frac{C_h L^{\frac{3}{2}}}{\sqrt{3}} \right) \right) V(t).
\end{aligned} \tag{3.19}$$

Thus, choose

$$\mu^* = \frac{1}{2} \max_{x \in [0, L]} \beta''(x) + C^{*2} \check{\epsilon}^2 \nu^2 + c\nu \left(1 + \frac{C_h L^{\frac{3}{2}}}{\sqrt{3}} \right), \tag{3.20}$$

if μ verifies $\mu \geq \mu^* + \varepsilon$ for some $\varepsilon > 0$, one has

$${}_0^C D_t^\alpha V(t) \leq -2\varepsilon V(t). \tag{3.21}$$

Moreover, similar to the methods in [5], we set

$$\theta(t) = -2\varepsilon V(t) - {}_0^C D_t^\alpha V(t) \geq 0. \tag{3.22}$$

Since the Laplace transformation of function $\rho^2(\cdot, t)$ exists, denote $\hat{V}(s) := \int_0^\infty e^{-st} V(t) dt$ the Laplace transformation of function $V(t)$, one gets that

$$\hat{\theta}(s) = -2\varepsilon \hat{V}(s) - s^\alpha \hat{V}(s) + s^{\alpha-1} V(0). \tag{3.23}$$

So, $\hat{V}(s) = \frac{s^{\alpha-1} V(0) - \hat{\theta}(s)}{s^\alpha + 2\varepsilon}$. It therefore follows from the uniqueness and existence property of fractional-order equation (3.22), positivity of $\theta(t)$ that

$$V(t) = E_\alpha(-2\varepsilon t^\alpha) V(0) - \theta(t) * [t^{\alpha-1} E_{\alpha, \alpha}(-2\varepsilon t^\alpha)] \leq E_\alpha(-2\varepsilon t^\alpha) V(0). \tag{3.24}$$

Here $*$ denotes the convolution operator,

$$E_{\alpha, \beta}(t) = \sum_{k=0}^{\infty} \frac{t^k}{\Gamma(\alpha k + \beta)} \tag{3.25}$$

with $\alpha > 0$, $\beta > 0$ represents the famous two parameters Mittag-Leffler function and $E_\alpha(t) = E_{\alpha, 1}(t)$ is written in simplified form [33].

Recall that $\|\rho_0\|_{L^2(0, L)} \leq \left(1 + C_h \sqrt{L} \right) \|\omega_0\|_{L^2(0, L)}$. We have

$$V(t) \leq \frac{1}{2} \left(1 + C_h \sqrt{L} \right)^2 E_\alpha(-2\varepsilon t^\alpha) \|\omega_0\|_{L^2(0, L)}^2 \tag{3.26}$$

and then, get that

$$\|\omega(\cdot, t)\|_{L^2(0, L)} \leq \nu \left(1 + C_h \sqrt{L} \right) E_\alpha^{\frac{1}{2}}(-2\varepsilon t^\alpha) \|\omega_0\|_{L^2(0, L)}. \tag{3.27}$$

Thus, the studied system (1.1) under triggering condition (3.6) with event-triggered boundary controller designed as in (3.7) is Mittag-Leffler stable in Hilbert space $L^2(0, L)$ and moreover, $\sup_{t \geq 0} \|\omega(\cdot, t)\|_{L^2(0, L)} \leq r$ is satisfied for some $r = v \left(1 + C_h \sqrt{L}\right) r_0$ by noting that $E_a^{\frac{1}{2}}(-2\epsilon t^\alpha) \leq 1$ for all $t \geq 0$ provided that $\epsilon > 0$. These finish our proof. $\square$

3.3. Avoidance of the Zeno behavior

To implement above designed event-triggered boundary controller, for any interval $[0, T]$, the inter-event time defined by

$$T_k^* = \inf \{t_{k+1} - t_k\} \text{ for all } k = 0, 1, 2, \dots \quad (3.28)$$

with t_k determined by Eq.(3.6) must be lower bounded ensuring that the Zeno phenomenon can be avoided. That is, some constant $T_{\min} > 0$ can be found such that

$$T_k^* \geq T_{\min}, \quad \forall k = 0, 1, 2, \dots \quad (3.29)$$

To this end, we give our result as follows.

Theorem 3.2. *The inter-event time T_k^* defined as in (3.28) is lower bounded provided that all conditions presented in the statements of Theorem 3.1 are fulfilled.*

Proof. Let us define $\omega(x, t) - \omega(x, t_k) := \delta(x, t)$. By Eq. (1.2), one gets that $\delta(x, t)$ satisfies

$$\begin{aligned} {}^C_0 D_t^\alpha \delta(x, t) &= {}_0 I_t^{1-\alpha} \frac{\partial}{\partial t} \delta(x, t) = {}^C_0 D_t^\alpha \omega(x, t) \\ &= \beta(x) \frac{\partial^2 \delta(x, t)}{\partial x^2} + \beta(x) \frac{\partial^2 \omega(x, t_k)}{\partial x^2} + f(\delta(x, t) + \omega(x, t_k), x, t) \end{aligned} \quad (3.30)$$

subject to the condition $\delta(x, t_k) = 0$ and the BCs

$$\frac{\partial \delta}{\partial x}(0, t) = p\delta(0, t), \quad \frac{\partial \delta}{\partial x}(L, t) + q\delta(L, t) = 0, \quad t \in [t_k, t_{k+1}). \quad (3.31)$$

Notice that no control update is required when $\|\omega(\cdot, t_k)\|_{L^2(0, L)} = 0$, before giving the proof, we claim that given any $\epsilon > 0$, the designed triggering condition (3.6) can be always violated for some $t_{k+1} < \infty$ provided that $\|\omega(\cdot, t_k)\|_{L^2(0, L)} > 0$.

More precisely, we divide our analysis into two cases.

Case 1 when $\|\omega(\cdot, t)\|_{L^2(0, L)} \neq 0$ holds true for all $t > t_k$. Since system (1.1) is Mittag-Leffler stable, by inequality (3.27), one has

$$\frac{\|\delta(\cdot, t)\|_{L^2(0, L)}}{\|\omega(\cdot, t)\|_{L^2(0, L)}} \geq \frac{\|\omega(\cdot, t_k)\|_{L^2(0, L)}}{\|\omega(\cdot, t)\|_{L^2(0, L)}} - 1 \geq \frac{\|\omega(\cdot, t_k)\|_{L^2(0, L)}}{v\|\omega_0\|_{L^2(0, L)} \left(1 + C_h \sqrt{L}\right) E_a^{\frac{1}{2}}(-2\epsilon t^\alpha)} - 1 \rightarrow \infty \quad \text{as } t \rightarrow \infty. \quad (3.32)$$

Due to the fact that $\|\omega(\cdot, t)\|_{L^2(0, L)}$ is continuous and $\|\delta(\cdot, t_k)\|_{L^2(0, L)} = 0$, then some finite $t_{k+1} > t_k$ can be always found when the triggering condition (3.6) is violated.

Case 2 when a $t^\dagger > t_k$ can be found satisfying $\|\omega(\cdot, t^\dagger)\|_{L^2(0, L)} = 0$. By setting $t_{\inf} = \inf \{t^\dagger > t_k : \|\omega(\cdot, t^\dagger)\|_{L^2(0, L)} = 0\}$, we get that $\|\omega(\cdot, t_{\inf})\|_{L^2(0, L)} = 0$ and

$$\|\delta(\cdot, t_{\inf})\|_{L^2(0, L)} \geq \|\omega(\cdot, t_k)\|_{L^2(0, L)} - \|\omega(\cdot, t_{\inf})\|_{L^2(0, L)} > 0. \quad (3.33)$$

Moreover, by noticing that $\delta(\cdot, t_k) = 0$ and $\lim_{t \rightarrow t_{\inf}^-} \frac{\|\delta(\cdot, t)\|_{L^2(0, L)}}{\|\omega(\cdot, t)\|_{L^2(0, L)}} = \infty$, given any $\epsilon > 0$, an instant $t_{k+1} \in (t_k, t_{\inf})$ when proposed triggering condition (3.6) is violated will always exist.

Next, by the definition of semilinear function $f(\cdot, x, t)$, according to Lemma 2.1, we see that

$$\begin{aligned} \delta(x, t) &= \sum_{n=1}^{\infty} \int_{t_k}^t \frac{E_{\alpha, \alpha}(\lambda_n(t-\tau)^\alpha)}{(t-\tau)^{1-\alpha}} d\tau \left(\beta(\cdot) \frac{\partial^2 \omega(\cdot, t_k)}{\partial x^2}, \xi_n \right)_{L^2(0, L)} \xi_n(x) \\ &\quad - \sum_{n=1}^{\infty} \int_0^{t_k} k_n(t-\tau, t_k-\tau) d\tau \left(\beta(\cdot) \frac{\partial^2 \omega(\cdot, t_k)}{\partial x^2}, \xi_n \right)_{L^2} \xi_n(x) \\ &\quad + \sum_{n=1}^{\infty} \int_{t_k}^t \frac{E_{\alpha, \alpha}(\lambda_n(t-\tau)^\alpha)}{(t-\tau)^{1-\alpha}} (\hat{f}(\cdot, \tau), \xi_n)_{L^2(0, L)} d\tau \xi_n(x) \\ &\quad - \sum_{n=1}^{\infty} \int_0^{t_k} k_n(t-\tau, t_k-\tau) (\hat{f}(\cdot, \tau), \xi_n)_{L^2(0, L)} d\tau \xi_n(x), \end{aligned} \quad (3.34)$$

where

$$k_n(t-\tau, t_k-\tau) = \frac{E_{\alpha, \alpha}(\lambda_n(t_k-\tau)^\alpha)}{(t_k-\tau)^{1-\alpha}} - \frac{E_{\alpha, \alpha}(\lambda_n(t-\tau)^\alpha)}{(t-\tau)^{1-\alpha}} \quad (3.35)$$

and $\hat{f}(x, t) = f(\delta(x, t) + \omega(x, t_k), x, t)$ satisfies

$$\|\hat{f}(\cdot, t)\|_{L^2(0,L)} \leq c \|\omega(\cdot, t)\|_{L^2(0,L)} \leq c\nu r_0 \left(1 + C_h \sqrt{L}\right) \quad (3.36)$$

by following the inequality (3.27) and the fact that $E_{\alpha}^{\frac{1}{2}}(-2\varepsilon t^{\alpha}) \leq 1$, $\forall t \geq 0$ with $\varepsilon > 0$. Moreover, since Mittag-Leffler function $E_{\alpha,\alpha}(\lambda t^{\alpha})$ with $\lambda \leq 0$, $t \geq 0$ satisfies $E_{\alpha,\alpha}(\lambda t^{\alpha}) \leq 1$ and $\frac{d}{dt} E_{\alpha,\alpha}(\lambda t^{\alpha}) < 0$, one gets $k_n(t - \tau, t_k - \tau) \geq 0$, $\forall t \geq t_k$ holds true and

$$\begin{aligned} \delta(x, t) &\leq \int_{t_k}^t \frac{1}{(t-\tau)^{1-\alpha}} d\tau \left| \sum_{n=1}^{\infty} \left(\beta(\cdot) \frac{\partial^2 \omega(\cdot, t_k)}{\partial x^2}, \xi_n \right)_{L^2(0,L)} \xi_n(x) \right| \\ &\quad - \sum_{n=1}^{\infty} \int_0^{t_k} k_n(t - \tau, t_k - \tau) d\tau \left(\beta(\cdot) \frac{\partial^2 \omega(\cdot, t_k)}{\partial x^2}, \xi_n \right)_{L^2} \xi_n(x) \\ &\quad + \int_{t_k}^t \frac{1}{(t-\tau)^{1-\alpha}} \left| \sum_{n=1}^{\infty} (\hat{f}(\cdot, \tau), \xi_n)_{L^2(0,L)} \xi_n(x) \right| d\tau \\ &\quad - \sum_{n=1}^{\infty} \int_0^{t_k} k_n(t - \tau, t_k - \tau) (\hat{f}(\cdot, \tau), \xi_n)_{L^2(0,L)} d\tau \xi_n(x). \end{aligned} \quad (3.37)$$

It then follows from $k_n(t - \tau, t_k - \tau) \geq 0$,

$$\int_0^{t_k} k_n(t - \tau, t_k - \tau) d\tau = \frac{E_{\alpha}(\lambda_n t_k^{\alpha}) - E_{\alpha}(\lambda_n t^{\alpha}) + 1 - E_{\alpha}(\lambda_n (t - t_k)^{\alpha})}{\lambda_n} \quad (3.38)$$

that

$$\begin{aligned} \|\delta(\cdot, t)\|_{L^2(0,L)} &\leq \frac{\sqrt{L}}{\alpha} \left(\left\| \beta(\cdot) \frac{\partial^2 \omega(\cdot, t_k)}{\partial x^2} \right\|_{L^2} + c\nu r_0 \left(1 + C_h \sqrt{L}\right) \right) (t - t_k)^{\alpha} \\ &\quad + \sqrt{\sum_{n=1}^{\infty} \left(\int_0^{t_k} k_n(t - \tau, t_k - \tau) d\tau \right)^2 \left\| \beta(\cdot) \frac{\partial^2 \omega(\cdot, t_k)}{\partial x^2} \right\|_{L^2(0,L)}^2} \\ &\quad + \sqrt{\sum_{n=1}^{\infty} \left(\int_0^{t_k} k_n(t - \tau, t_k - \tau) \|\hat{f}(\cdot, \tau)\|_{L^2(0,L)} d\tau \right)^2} \\ &\leq M_k(t), \quad \forall t \geq t_k, \end{aligned} \quad (3.39)$$

where

$$M_k(t) = \frac{C_* \sqrt{L}}{\alpha} (t - t_k)^{\alpha} + C_{**} \sqrt{\sum_{n=1}^{\infty} \frac{\left(E_{\alpha}(\lambda_n t_k^{\alpha}) - E_{\alpha}(\lambda_n t^{\alpha}) + 1 - E_{\alpha}(\lambda_n (t - t_k)^{\alpha}) \right)^2}{\lambda_n^2}} \quad (3.40)$$

with $\lambda_n \leq 0$ given in Section 2 and $C_{**} = c\nu r_0 \left(1 + C_h \sqrt{L}\right) + \left\| \beta(\cdot) \frac{\partial^2 \omega(\cdot, t_k)}{\partial x^2} \right\|_{L^2(0,L)}$.

Further, notice that

$$M_k(t_k) = 0, \quad \frac{d}{dt} M_k(t) > 0, \quad \lim_{t \rightarrow \infty} M_k(t) = \infty \quad (3.41)$$

and $\|\delta(\cdot, t)\|_{L^2(0,L)}$ is an increasing function while $\|\omega(\cdot, t)\|_{L^2(0,L)}$ is decreasing. Based on the definition of triggering condition (3.6) and the continuity of $\|\omega(\cdot, t)\|_{L^2(0,L)}$, if some t_{k+1} which satisfies

$$M_k(t_{k+1}) = \varepsilon \|\omega(\cdot, t_{k+1})\|_{L^2(0,L)} \quad (3.42)$$

exists, a $t_{k+1}^* \geq t_{k+1}$ can then be found when our proposed triggering condition (3.6) is violated. In the sequel, we therefore, aim to discuss properties of the following continuous function:

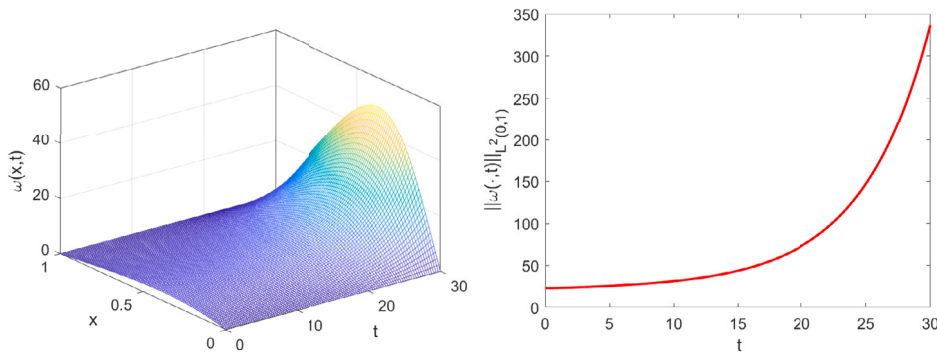
$$\psi(t) = \frac{M_k(t)}{\|\omega(\cdot, t)\|_{L^2(0,L)}^2} \quad (3.43)$$

by supposing that $\|\omega(\cdot, t)\|_{L^2(0,L)} > 0$ for all $t \in [t_k, t_{k+1})$. The reason is that when there exists a $t_{temp} \in (t_k, t_{k+1})$ yielding $\|\omega(\cdot, t_{temp})\|_{L^2(0,L)} = 0$, define $t^* = \inf\{t_{temp} > t_k : \|\omega(\cdot, t_{temp})\|_{L^2(0,L)} = 0\}$ and based on above arguments, the next instant t_{k+1}^* exists and satisfies $t_{k+1}^* \in (t_k, t_{*})$. This contradicts to the definition of t_{k+1} . Then, we suppose that $\|\omega(\cdot, t)\|_{L^2(0,L)} > 0$ holds true for all $t \in [t_k, t_{k+1})$.

More precisely, by setting $T_k = T_k(t) = t - t_k$, one has

$$\begin{aligned} \psi(T_k) &= \frac{M_k(T_k)}{\|\omega(\cdot, t_k + T_k)\|_{L^2(0,L)}^2} \\ &= \frac{C_* \sqrt{L} T_k^{\alpha}}{\alpha \|\omega(\cdot, t_k + T_k)\|_{L^2(0,L)}} + C_{**} \frac{\sqrt{\sum_{n=1}^{\infty} \frac{\left(E_{\alpha}(\lambda_n t_k^{\alpha}) - E_{\alpha}(\lambda_n (t_k + T_k)^{\alpha}) + 1 - E_{\alpha}(\lambda_n T_k^{\alpha}) \right)^2}{\lambda_n^2}}}{\|\omega(\cdot, t_k + T_k)\|_{L^2(0,L)}}. \end{aligned} \quad (3.44)$$

Consider that the function $\|\omega(\cdot, t_k + T_k)\|_{L^2(0,L)}$ is continuous and $M_k(0) = 0$, some $T_k^* = \min_{t \in [t_k, t_{k+1})} T_k(t)$, which leads to



(i) Solution of system (4.1) with $u \equiv 0$. (ii) L^2 -norm of the solution to system (4.1) with $u \equiv 0$.

Fig. 2. Evolution and L^2 -norm of the solution for system (4.1) under the condition that $u \equiv 0$.

$$\check{e} = \psi(T_k^*), \quad (3.45)$$

exists. Notice that T_k^* means the time $\psi(t)$ takes to go from 0 to $\check{e}$, then $T_k^* > 0$ should be uniformly positive.

Finally, we prove that for any interval $[0, T]$, the minimum of considered inter-event time defined as

$$T_{\min} := \min_{k=0,1,2,\dots} \{T_k^*\}, \quad (3.46)$$

for triggering instants $\{t_k\}_{k \geq 0}$ is lower bounded. If not, we see $T_{\min} = 0$ and get that some $T_k^* = 0$ or a subsequence, still written as $\{T_k^*\}_{k \geq 0}$, can be found satisfying

$$\sum_{k=0}^{\infty} T_k^* \leq T \text{ and } T_k^* \rightarrow 0 \text{ as } k \rightarrow \infty, \quad (3.47)$$

which contradicts to the definition of T_k^* . This is because that since $M_k(0) = 0$ and $M_k(t)$ is continuous, given any triggering threshold $\check{e} > 0$, if $T_k^* > 0$ goes to small enough, finding some $T_k^* > 0$ for $\psi(t)$ to go from 0 to $\check{e}$ is impossible. Hence, we claim that the inter-event time T_k^* defined as in (3.28) is lower bounded. These finish our proof. $\square$

4. Numerical example

To illustrate our results, let us consider the following example:

$$\begin{cases} {}^C_0 D_t^{0.75} \omega(x, t) = 0.5 (0.1 + 0.64(x - 0.1)^2)^2 \frac{\partial^2 \omega(x, t)}{\partial x^2} + 1.28 (\sqrt{4\omega^2(x, t) + 0.81} - 0.9) - 2\omega(x, t), \\ (x, t) \in (0, 1) \times [0, \infty), \\ \begin{cases} \frac{\partial \omega}{\partial x}(0, t) - 3\omega(0, t) = 0, \\ \frac{\partial \omega}{\partial x}(1, t) + 2\omega(1, t) = u(t), \end{cases} t \in [0, \infty), \\ \omega(x, 0) = 20x(1 - x)e^{-x}, x \in (0, 1). \end{cases} \quad (4.1)$$

So, $\alpha = 0.75$, $L = 1$, $p = 3$, $q = 2$, $\beta(x) = 0.5 (0.1 + 0.64(x - 0.1)^2)^2$ and $f(\omega, x, t) = 1.28 (\sqrt{4\omega^2 + 0.81} - 0.9) - 2\omega$, $\omega_0(x) = 20x(1 - x)e^{-x}$. Clearly, Assumption A_0 is satisfied for $c = 0.56$. With these, we obtain Fig. 2, which indicates that when $u \equiv 0$, system (4.1) is unstable. In the sequel, we focus on testing our theoretical results by implementing them on the system (4.1).

To this end, by $\beta(x) = 0.5 (0.1 + 0.64(x - 0.1)^2)^2$, one gets that $\frac{3}{16} \frac{\beta'^2(x)}{\beta(x)} - \frac{\beta''(x)}{4} = -0.032$ is a constant, the conditions (3.8) hold true and one has $\beta_{\min} = 0.005$, $\beta_{\max} = 0.1912$, $\max_{x \in [0,1]} \beta''(x) = 2.12$. Choose $\mu = 3.5$ in the target system (3.3), we see [34]

$$h(x, \zeta) = -d^* \bar{x} \frac{\beta^{1/4}(x)}{\beta^{3/4}(\zeta)} \frac{I_1(\sqrt{d^*(\bar{x}^2 - \zeta^2)})}{\sqrt{d^*(\bar{x}^2 - \zeta^2)}} + \frac{pd^*}{\sqrt{d^* + p^2}} \frac{\beta^{1/4}(x)}{\beta^{3/4}(\zeta)} \int_0^{\bar{x} - \zeta} \sinh\left(\frac{\sqrt{d^* + p^2}}{2} s\right) \frac{I_0(\sqrt{d^*(\bar{x} + \zeta)(\bar{x} - \zeta - s)})}{e^{ps/2}} ds, \quad (4.2)$$

where $d^* = \frac{3.5}{\sqrt{\beta(0)}} \approx 15.174$, $p = 3$ and $\bar{x} = \eta(x)$, $\bar{\zeta} = \eta(\zeta)$ with

$$\eta(s) = 1.258 (\arctan(0.8(s - 0.1)) + \arctan(0.08)). \quad (4.3)$$

From the Theorem 2 of [35], one gets that $C_h = Ne^{2N}$ with $N = 0.5(1 + e^{-3})$. Set $\check{e} = 0.15$ in (3.6) and $\mu^* = 3.3$, $\varepsilon = 0.15$, we then, get that all conditions presented in the statements of Theorem 3.1 are fulfilled.

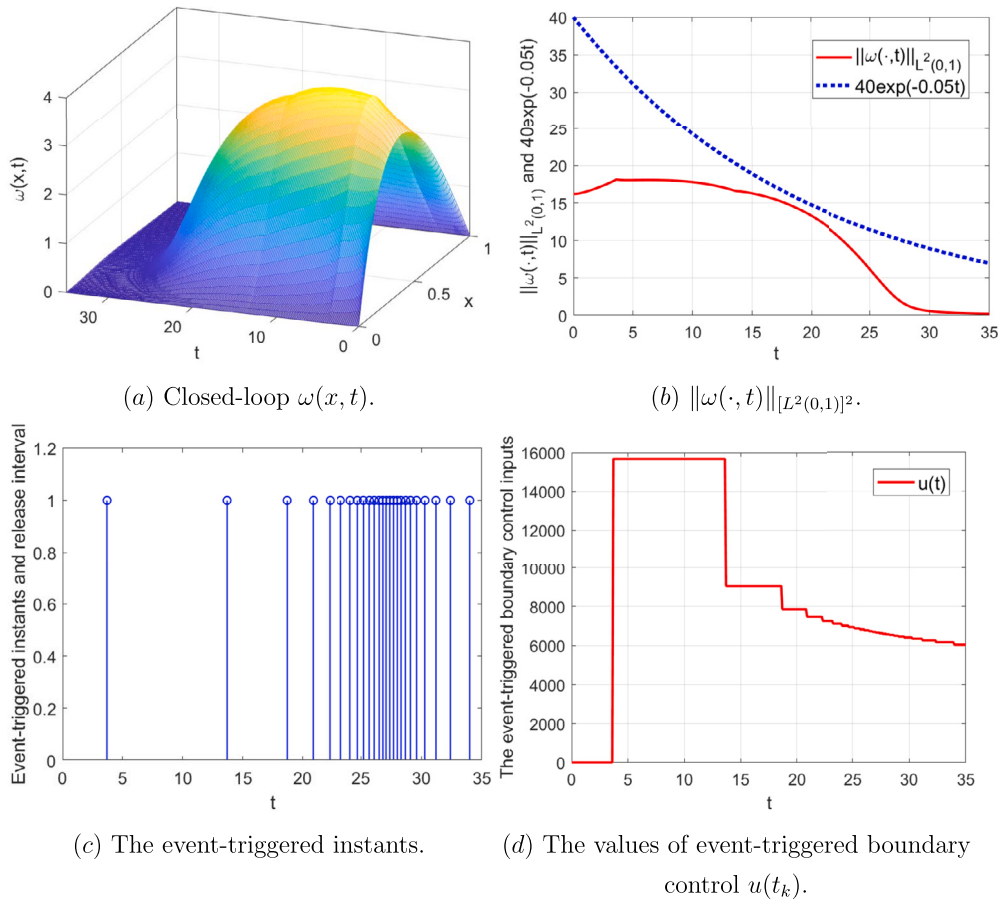


Fig. 3. Plot of solution for closed-loop system (4.1).

As a consequence, solution of the considered system (4.1) within $t \in [0, 35]$ can be calculated, which indicates that $T_k^* = 0.2$. For a numerical illustration of our results, please see Fig. 3 (a) – (d), where evolution of the solution, solution in the sense of L^2 -norm, values of proposed event-triggered control inputs $u(t)$ and the corresponding triggering instants t_k are plotted, respectively. Notice that system (4.1) is Mittag-Leffler stable and the T_k^* that are calculated by following the triggering condition $\tilde{e} = \left\| \frac{\omega(\cdot, t_k)}{\omega(\cdot, t_k + T_k^*)} - 1 \right\|$ increases as $t_k \rightarrow \infty$. We obtain that the $T_{\min}$ for our designed event-triggered boundary control scheme is still 0.2. All these illustrations yield that our designed event-triggered boundary control strategy can yield an effective and efficient performance.

5. Conclusions

This paper is devoted to designing an event-triggered controller using backstepping to Mittag-Leffler stabilize the boundary controlled semilinear time-fractional diffusion systems with spatially-varying coefficients. Event-triggered Mittag-Leffler boundary feedback stabilization of the considered systems and the proof for guaranteeing non-occurrence of the Zeno phenomenon are presented. At last, we work out a numerical simulation to test the effectiveness and efficiency of our results. This is very appealing in those science and engineering digital applications. Moreover, we claim that the studies of event-triggered control problems of fractional-order PDE systems are still at their early stage. Various open questions such as proposing more advanced event-triggered boundary or in-domain controller design methods for more complex nonlinear fractional-order PDE systems as well as the case of only knowing partial state variables, etc. are of great interest.

Data availability

No data was used for the research described in the article.

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Appendix A

A.1. Proof of Proposition 3.1

The goal of this proof is to prove the equivalent between system (3.1) and system (3.3) via integral transformation (3.2) under these conditions of Proposition 3.1. Toward this aim, we split our proof into two steps as follows.

Step 1. System (3.4) has a twice continuously differentiable and unique solution over the domain D . Define an operator $\mathcal{H}$ as follows

$$(\mathcal{H}\omega)(x, t) := \rho(x, t) = \omega(x, t) - \int_0^x h(x, \zeta) \omega(\zeta, t) d\zeta, \quad (\text{A.1})$$

we shall show that both $\mathcal{H}$ and its inverse are bounded, i.e., operator $\mathcal{H}$ is invertible.

To begin with, by the second and the last equation of (3.4), one has

$$h(x, x) = \frac{-1}{2} \int_0^x \frac{\mu}{\sqrt{\beta(x)\beta(s)}} ds. \quad (\text{A.2})$$

Next, we simplify the hyperbolic PDE (3.4) by introducing the change of variables as follows:

$$\hat{h}(\bar{x}, \bar{\zeta}) = h(x, \zeta) \beta(\zeta) \text{ and } \bar{\zeta} = \eta(\zeta), \quad \bar{x} = \eta(x) \quad (\text{A.3})$$

with $\eta(x) = \sqrt{\beta(0)} \int_0^x (\beta(s))^{-1/2} ds$, which yield that

$$\begin{cases} \frac{\partial^2 \hat{h}(\bar{x}, \bar{\zeta})}{\partial \bar{x}^2} - \frac{\partial^2 \hat{h}(\bar{x}, \bar{\zeta})}{\partial \bar{\zeta}^2} = \frac{\mu}{\beta(0)} \hat{h}(\bar{x}, \bar{\zeta}) - \frac{\beta'(\zeta)}{2\sqrt{\beta(0)\beta(\zeta)}} \frac{\partial \hat{h}(\bar{x}, \bar{\zeta})}{\partial \bar{\zeta}} + \frac{\beta'(x)}{2\sqrt{\beta(0)\beta(x)}} \frac{\partial \hat{h}(\bar{x}, \bar{\zeta})}{\partial \bar{x}}, \\ \frac{\partial \hat{h}}{\partial \bar{\zeta}}(\bar{x}, 0) = p \hat{h}(\bar{x}, 0), \\ \hat{h}(\bar{x}, \bar{x}) = \frac{-\mu \bar{x}}{2\sqrt{\beta(0)}} \sqrt{\beta(\eta^{-1}(\bar{x}))}. \end{cases} \quad (\text{A.4})$$

By letting $\bar{h}(\bar{x}, \bar{\zeta}) = \hat{h}(\bar{x}, \bar{\zeta}) / \sqrt{\beta(x)\beta(\zeta)}$, we further obtain that above hyperbolic PDE (3.4) is equivalent to

$$\begin{cases} \frac{\partial^2 \bar{h}(\bar{x}, \bar{\zeta})}{\partial \bar{x}^2} - \frac{\partial^2 \bar{h}(\bar{x}, \bar{\zeta})}{\partial \bar{\zeta}^2} = \frac{\theta(x)}{\beta(0)} \bar{h}(\bar{x}, \bar{\zeta}), \\ \frac{\partial \bar{h}}{\partial \bar{\zeta}}(\bar{x}, 0) = \left(p - \frac{\beta'(0)}{4\beta(0)}\right) \bar{h}(\bar{x}, 0), \\ \bar{h}(\bar{x}, \bar{x}) = \frac{-\mu \bar{x}}{2\sqrt{\beta(0)}}, \end{cases} \quad (\text{A.5})$$

with $\theta(x) = \mu + \frac{3}{16} \left(\frac{\beta'^2(x)}{\beta(x)} - \frac{\beta'^2(\zeta)}{\beta(\zeta)} \right) - \frac{\beta''(x) - \beta''(\zeta)}{4}$. Moreover, since $\frac{3}{16} \frac{\beta'^2(x)}{\beta(x)} - \frac{\beta''(x)}{4} = C$ is supposed to be a constant, it is easy to see that $\frac{\theta(x)}{\beta(0)}$ is also a constant. Therefore, based on the statements for hyperbolic PDE (108-110) in [35], one sees that PDE (3.4) admits a twice continuously differentiable and unique solution given by

$$h(x, \zeta) = -d^* \bar{x} \frac{\beta^{1/4}(x)}{\beta^{3/4}(\zeta)} \frac{I_1\left(\sqrt{d^*(\bar{x}^2 - \bar{\zeta}^2)}\right)}{\sqrt{d^*(\bar{x}^2 - \bar{\zeta}^2)}} + \frac{pd^*}{\sqrt{d^* + p^2}} \frac{\beta^{1/4}(x)}{\beta^{3/4}(\zeta)} \int_0^{\bar{x} - \bar{\zeta}} \sinh\left(\frac{\sqrt{d^* + p^2}}{2} s\right) \frac{I_0\left(\sqrt{d^*(\bar{x} + \bar{\zeta})(\bar{x} - \bar{\zeta} - s)}\right)}{e^{ps/2}} ds. \quad (\text{A.6})$$

Here $d^* = \frac{\mu}{\sqrt{\beta(0)}}$ is a constant, $\bar{x}, \bar{\zeta}, \eta(x)$ are defined as in (A.3) and I_0, I_1 are modified Bessel function of zero order and first order. Further, similar to the method used for proving the Proposition 1 of [5], one gets that both $\mathcal{H}$ and its inverse operator are bounded, i.e., transformation (3.2) is invertible.

Step 2. Equivalent transformation between system (3.1) and system (3.3) via integral transformation (3.2) under the given conditions.

With these preparations, consider the system (3.3) and the integral transformation (3.2), it yields that

$$\begin{aligned} 0 &= {}_0^C D_t^\alpha \rho(x, t) - \beta(x) \frac{\partial^2 \rho(x, t)}{\partial x^2} + \mu \rho(x, t) - f(\omega(x, t), x, t) + \int_0^x h(x, \zeta) f(\omega(\zeta, t), \zeta, t) d\zeta \\ &= {}_0^C D_t^\alpha \omega(x, t) - \int_0^x h(x, \zeta) {}_0^C D_t^\alpha \omega(\zeta, t) d\zeta - \beta(x) \frac{\partial^2 \omega(x, t)}{\partial x^2} + \beta(x) \frac{\partial^2 \left(\int_0^x h(x, \zeta) \omega(\zeta, t) d\zeta \right)}{\partial x^2} \\ &\quad + \mu \omega(x, t) - \mu \int_0^x h(x, \zeta) \omega(\zeta, t) d\zeta - f(\omega(x, t), x, t) + \int_0^x h(x, \zeta) f(\omega(\zeta, t), \zeta, t) d\zeta. \end{aligned} \quad (\text{A.7})$$

Since

$$\frac{\partial^2 \left(\int_0^x h(x, \zeta) \omega(\zeta, t) d\zeta \right)}{\partial x^2} = \frac{d}{dx} h(x, x) \omega(x, t) + h(x, x) \frac{\partial \omega(x, t)}{\partial x} + \int_0^x \frac{\partial^2 h(x, \zeta)}{\partial x^2} \omega(\zeta, t) d\zeta + \frac{\partial h(x, x)}{\partial x} \omega(x, t) \quad (\text{A.8})$$

and

$$\begin{aligned} \int_0^x h(x, \varsigma) D_t^\alpha \omega(\varsigma, t) d\varsigma &= \beta(x) h(x, x) \frac{\partial \omega(x, t)}{\partial x} - h(x, 0) \beta(0) \frac{\partial \omega(0, t)}{\partial x} - \beta(x) \frac{\partial h(x, x)}{\partial \varsigma} \omega(x, t) \\ &\quad - \frac{d}{dx} \beta(x) h(x, x) \omega(x, t) + \frac{d}{dx} \beta(0) h(x, 0) \omega(0, t) + \beta(0) \frac{\partial h(x, 0)}{\partial \varsigma} \omega(0, t) \\ &\quad + \int_0^x \frac{\partial^2 [h(x, \varsigma) \beta(\varsigma)]}{\partial \varsigma^2} \omega(\varsigma, t) d\varsigma + \int_0^x h(x, \varsigma) f(\omega(\varsigma, t), \varsigma, t) d\varsigma, \end{aligned} \quad (\text{A.9})$$

by $\frac{d}{dx} h(x, x) = \frac{\partial h(x, x)}{\partial x} + \frac{\partial h(x, x)}{\partial \varsigma}$, it follows from Eq. (A.7) and the system (3.1) that

$$\begin{aligned} 0 &= {}^C D_t^\alpha \omega(x, t) - \int_0^x h(x, \varsigma) \beta(\varsigma) \frac{\partial^2 \omega(\varsigma, t)}{\partial \varsigma^2} d\varsigma - \int_0^x h(x, \varsigma) f(\omega(\varsigma, t), \varsigma, t) d\varsigma \\ &\quad - \beta(x) \frac{\partial^2 \omega(x, t)}{\partial x^2} + \beta(x) \frac{d}{dx} h(x, x) \omega(x, t) + \beta(x) h(x, x) \frac{\partial \omega(x, t)}{\partial x} \\ &\quad + \int_0^x \beta(x) \frac{\partial^2 h(x, \varsigma)}{\partial x^2} \omega(\varsigma, t) d\varsigma + \beta(x) \frac{\partial h(x, x)}{\partial x} \omega(x, t) + \mu \omega(x, t) \\ &\quad - \mu \int_0^x h(x, \varsigma) \omega(\varsigma, t) d\varsigma - f(\omega(x, t), x, t) + \int_0^x h(x, \varsigma) f(\omega(\varsigma, t), \varsigma, t) d\varsigma \\ &= h(x, 0) \beta(0) \frac{\partial \omega(0, t)}{\partial x} - \beta(0) \frac{\partial h(x, 0)}{\partial \varsigma} \omega(0, t) - h(x, 0) \frac{d\beta(0)}{dx} \omega(0, t) \\ &\quad + 2\beta(x) \frac{d}{dx} h(x, x) \omega(x, t) + h(x, x) \frac{d\beta(x)}{dx} \omega(x, t) + \mu \omega(x, t) \\ &\quad + \int_0^x \beta(x) \frac{\partial^2 h(x, \varsigma)}{\partial x^2} \omega(\varsigma, t) d\varsigma - \int_0^x \frac{\partial^2 (h(x, \varsigma) \beta(\varsigma))}{\partial \varsigma^2} \omega(\varsigma, t) d\varsigma - \mu \int_0^x h(x, \varsigma) \omega(\varsigma, t) d\varsigma \\ &= \left(2\beta(x) \frac{d}{dx} h(x, x) + h(x, x) \frac{d\beta(x)}{dx} + \mu \right) \omega(x, t) \\ &\quad + \int_0^x \left(\beta(x) \frac{\partial^2 h(x, \varsigma)}{\partial x^2} - \frac{\partial^2 [h(x, \varsigma) \beta(\varsigma)]}{\partial \varsigma^2} - \mu h(x, \varsigma) \right) \omega(\varsigma, t) d\varsigma \\ &\quad + h(x, 0) \beta(0) \frac{\partial \omega(0, t)}{\partial x} - \beta(0) \frac{\partial h(x, 0)}{\partial \varsigma} \omega(0, t) - h(x, 0) \frac{d\beta(0)}{dx} \omega(0, t). \end{aligned} \quad (\text{A.10})$$

Further, the boundary conditions of systems (3.1), (3.3) yield that if we set $\hat{p} = p$ and $\hat{q} = -h(L, L) + q$, it leads to $h(0, 0) = 0$. Then, the hyperbolic PDE (3.4) on kernel function $h(x, \varsigma)$ should be fulfilled. Finally, according to the results obtained in Step 1, we see the equivalence between dynamic systems (3.1) and (3.3). The proof is complete.

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